# Computer Networks-II

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# TCP/IP(Transmission Control Protocol/Internet Protocol)

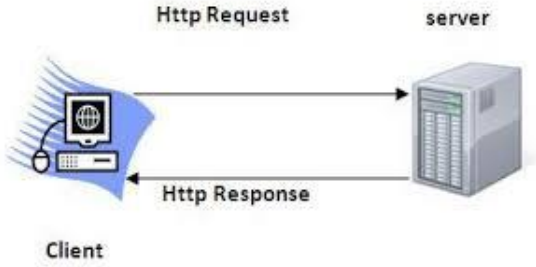
- TCP/IP, or Transmission Control Protocol/Internet Protocol, is a suite of communication protocols used to interconnect network devices on the internet. TCP/IP can also be used as a communications protocol in a private computer network (an intranet or extranet).
- TCP defines how applications can create channels of communication across a network. It also manages how a message is assembled into smaller packets before they are transmitted over the internet and reassembled in the right order at the destination address.
- IP defines how to address and route each packet to make sure it reaches the right destination.
- HTTP,FTP,HTTPS are some protocols that use TCP/IP

# FTP(File Transfer Protocol)

- File Transfer Protocol (FTP) is a standard Internet protocols for transmitting files between computers on the Internet over TCP/IP connections.
- FTP is a client-server protocol where a client will ask for a file, and a local or remote server will provide it.
- FTP relies on two communications channels between client and server: a command channel for controlling the conversation and a data channel for transmitting file content.
- A user typically needs to log on to the FTP server, although some servers make some or all of their content available without login, known as anonymous FTP
- PORT NO:20(Data connection) and 21(Control Connection)

# HTTP(Hyper Text Transfer Protocol)

- Basically, HTTP is a TCP/IP based communication protocol, that is used to deliver data (HTML files, image files, query results, etc.) on the World Wide Web.
- The default port is TCP 80.
- HTTP functions as a request-response protocol in the client–server computing model.
- A web browser, for example, may be the *client* and an application running on a computer hosting a website may be the *server*. The client submits an HTTP *request* message to the server.
- The server, which provides *resources* such as HTML files and other content, or performs other functions on behalf of the client, returns a *response* message to the client. The response contains completion status information about the request and may also contain requested content in its message body.



# SMTP(Simple Mail Transfer Protocol)

- SMTP stands for Simple Mail Transfer Protocol.
- It provides mail exchange between users on the same or different computers.
- The main purpose of SMTP is used to set up communication rules between servers.
- SMTP also have a way of handling the errors such as incorrect email address. For example, if the recipient address is wrong, then receiving server reply with an error message of some kind.
- PORT NO:25

# POP(Post Office Protocol)

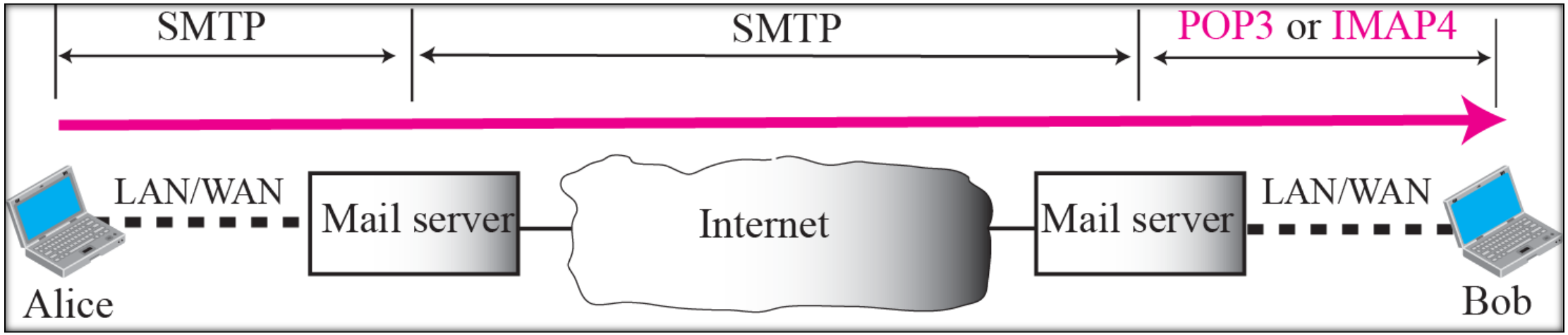
- Post Office Protocol (POP) is a type of computer networking and Internet standard protocol that extracts and retrieves email from a remote mail server for access by the host machine.
- POP uses the TCP/IP protocol stack for network connection and works with Simple Mail Transfer Protocol (SMTP) for end-to-end email communication, where POP pulls messages from the server to receiver's inbox and SMTP pushes them to the sender's server.
- POP is a protocol used to retrieve e-mail from a mail server.
- PORT NO:110 or 995

# IMAP(Internet Message Access Protocol)

- IMAP is the better option - and the recommended option - when you need to check your emails from multiple devices, such as a work laptop, a home computer, or a tablet, smartphone, or other mobile device. Tap into your synced (updated) account from any device with IMAP.

- POP3 downloads email from a server to a single computer, then deletes it from the server. Because your messages get downloaded to a single computer or device and then deleted from the server, it can appear that mail is missing or disappearing from your Inbox if you try to check your mail from a different computer.

# SMTP and POP3/IMAP



# Telnet

- It is an internet utility that lets you login to a remote computer system.
- To run a telnet session, you first have to run Telnet client and then connect to the desired Telnet site.

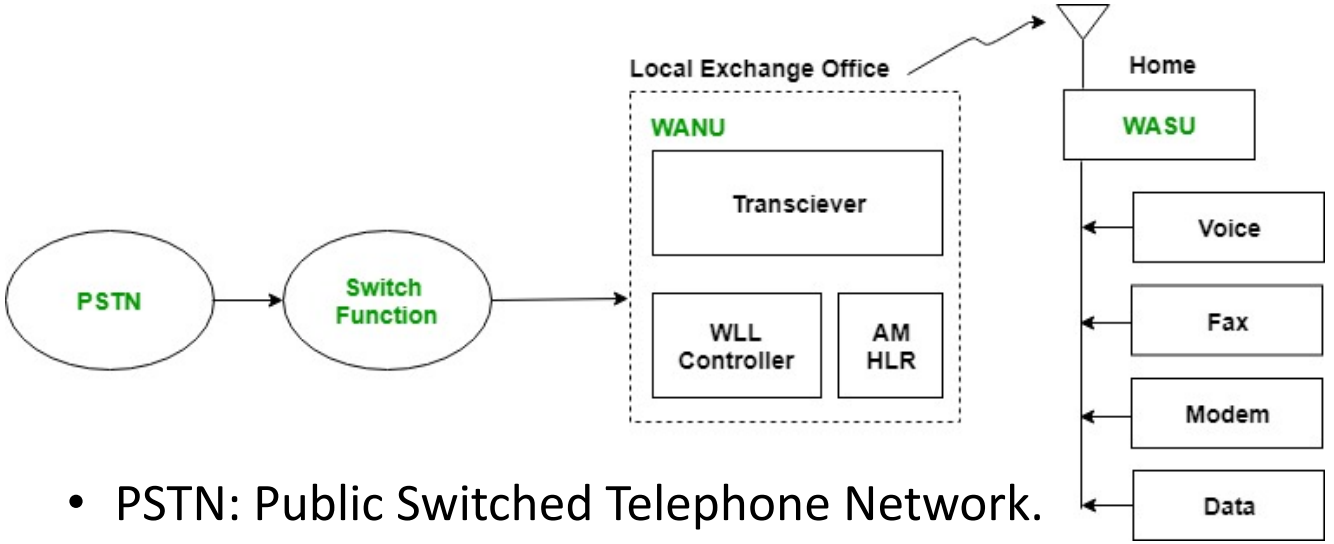
1. Run Telnet Client: Click at Start->Run command and then type Telnet.
2. Connect to telnet site: Once you have telnet client running on your system, you need to connect to telnet site. Click on Connect->Remote System in telnet window and then by specifying the telnet site address in which Host Name Box and then by clicking at Connect Button.
3. Start Browsing: Now you'll see default  menu of telnet site alongwith directions to use. Just type what you want to search in the search box.
4. Then after using Disconnect.

# GSM(Global System for Mobile Communication)

- It is a standard protocol for mobile communication.
- It uses TDMA(Time Division Multiple Access) which allows eight simultaneous calls on the same frequency. TDMA divides frequency channel into various time slots and then allocates these slots to multiple users. In this way, a single frequency can be used to support multiple, simultaneous data channels.
- GSM uses SIM(Subscriber Identification Module) to identify each user uniquely.

# WLL(Wireless in Local Loop)

- Local loop is a circuit line from a subscriber's phone to the local central office (LCO). But the implementation of local loop of wires is risky for the operators, especially in rural and remote areas due to less number of users and increased cost of installation. Hence, the solution for it is the usage of wireless local loop (WLL) which uses wireless links rather than copper wires to connect subscribers to the local central office.




- PSTN: Public Switched Telephone Network.
- WANU: Wireless Access Network Unit.
- AM HLR: Access Manager Home Location Register.
- WASU: Wireless Access Subscriber Unit

# WLL(Wireless in Local Loop)

- **Advantages of WLL:**
  - It eliminates the first mile or last mile construction of the network connection.
  - Low cost due to no use of conventional copper wires.
  - Much more secure due to digital encryption techniques used in wireless communication.
  - Highly scalable as it doesn’t require the installation of more wires for scaling it.
- **Features of WLL:**
  - Internet connection via modem
  - Data service
  - Voice service
  - Fax service

# GPRS(General Packet Radio Service)

- GPRS is a technology for radio transmission of small packets of data between mobile devices and internet.
- With this service you can access Internet, send emails, large data, watch real time News, movies and download games.
- Uses high speed data transfer typically between 56Kbits per second to 110Kbits.

# SMS, CHAT and Video Conferencing

- **SMS-Short Message Service-**The character limit for a single SMS message is 160 characters, however most modern phones and networks support concatenation and segment and rebuild messages up to 1600 characters.
- **VoIP (Voice over Internet Protocol)** is a technology that transmits voice as digital data over the internet, enabling phone, video, and messaging communication without traditional phone lines.
- **Video Conferencing:** It allows users to interact with each other face to face over internet. It allows audio and video to be sent over network. Protocol used is SIP(Session Initiation Protocol)

# URL and DNS

- URL is uniform resource locator also known as internet or web address.
- Eg: https://www.freedom-international.in/
- The "http" stands for Hypertext Transfer Protocol.  It let's the browser to know which protocol it is going to use to access the information specified in the domain. After the http or https is the colon ( : ) and two forward slashes ( // ) that separate the protocol from the remainder of the URL.
- Next, "www" stands for World Wide Web and is used to distinguish the content. This portion of the URL is not required and many times can be left out. This portion of the address can also be substituted for an important sub page known as a subdomain.

# URL and DNS

- Next, " freedom-international.in " is the domain name for the website. The last portion of the domain is known as the domain suffix. It is used to identify the type or location of the website. For example, ".com" is short for commercial, ".org" is short for an organization, “.in” for india and ".co.uk" is the United Kingdom. There are several domain suffixes available. To get a domain, you would register the name through a domain registrar.

# HTML and XML

| HTML(HyperText Markup Language) | XML(eXtensible Markup Language) |
|---|---|
| HTML is used **to display data** and focuses on how data looks. | XML is a software and hardware independent tool used **to transport and store data**. It focuses on what data is. |
| HTML is a **markup language** itself. | XML provides a **framework to define markup languages**. |
| HTML is **not case sensitive**. | XML is **case sensitive**. |
| HTML is a presentation language. | XML is neither a presentation language nor a programming language. |
| HTML **has its own predefined tags**. | You **can define tags according to your need**. |
| In HTML, it is **not necessary to use a closing tag**. | XML **makes it mandatory to use a closing tag**. |
| HTML is **static** because it is used to display data. | XML is **dynamic** because it is used to transport data. |
| HTML **does not preserve whitespaces**. | XML **preserve whitespaces**. |

# DHTML(Dynamic HTML)

- DHTML is a TERM used to describe the technologies used to make web pages dynamic and interactive.
- DHTML refers to web content that changes each time it is viewed. For example, the graphic can move from one location to another, in response to a user action, such as a mouse click.
- Enable a web page to react to user input without sending a request to a web server.
- Used to describe the combination of HTML, style sheets, and scripts that allow documents to be animate.

## Differences between XML and DHTML:

- DHTML is used to position information in a web page, and XML is used to describe that information.
- DHTML is HTML with JavaScript actions, whereas XML is more of universal way to transport info than a markup language.
- DHTML is used to display the dynamic web site pages, whereas XML is a markup language designed especially for Web documents.
- XML is an extensible markup language that was developed to retain the flexibility and power of HTML while reducing most of the complexity.

# Web Browsers and Web Servers

| | Web Browser | Web Server |
|---|---|---|
| **Purpose** | Web Browser is a software which is used to browse and display pages available over internet. | Web server is a software which provides these documents when requested by web browsers. |
| **Process** | A web browser sends request to server for web based documents and services. | Web server sees and approves those requests made by web browsers and sends the document in response. |
| **Process** | Web browser sends an HTTP Request and gets a HTTP Response. | Web server receives HTTP Request and sends a HTTP Response. |
| **Processing Model** | Web browser has no processing model. | Web server follows three major processing models: process based, thread based or hybrid. |
| **Data Storage** | Web browsers stores user data in cookies in local machine. | Web server provide an area to store the website. |
| **Installation** | Web Browser is installed on user's machine. | Web server can be installed any where but it need to be on a network or on local computer. |
| **Example** | Google Chrome | Apache Server |

# Web Hosting

- Web hosting is a service that allows organizations and individuals to post a website or web page onto the Internet.
- A web host, or web hosting service provider, is a business that provides the technologies and services needed for the website or webpage to be viewed in the Internet.
- Websites are hosted, or stored, on special computers called servers. When Internet users want to view your website, all they need to do is type your website address or domain into their browser.
- Their computer will then connect to your server and your webpages will be delivered to them through the browser.

# Web Hosting

- **Shared Hosting:** Web hosting providers usually offer more than one plan for each type of hosting. This type of hosting is the most common answer for most web hosting needs and it's an excellent solution for most small businesses and personal blogs. With this type of hosting, you're sharing one server with other clients. Websites hosted on the same server share all its resources, such as memory, computing power, disk space, and others.
- **Cloud Hosting:** Cloud hosting is currently the most reliable solution on the market. With cloud hosting, your host provides you with a cluster of servers — your files and resources are replicated on each server. When one of the cloud servers is busy or encounters a problem, your traffic is automatically routed to another server in the cluster. This results in little to no downtime, which is excellent if you own a very busy website.

# Web Hosting

- **Dedicated Hosting:** Dedicated hosting means that you have your own physical server that’s dedicated solely to your website. Therefore, you’re given incredible flexibility over how you want to manage your website. You can configure your server as you wish, choose the operating system and software you want to use, and set up the whole hosting environment according to your own needs.

# Web Scripting-Client Side & Server Side

| BASIS OF COMPARISON | SERVER-SIDE SCRIPTING | CLIENT-SIDE SCRIPTING |
|---|---|---|
| Use | It is used at the backend where the source code is not viewable or hidden at the client side (browser). | It is used at the front end which users can see from the browser. |
| Script Running | A web server runs the script for server-side scripting that creates the page which needs to be sent to the browser. | A browser runs the script for client-side scripting that is already present in the user's computer. |
| Occurrence | It happens when a user's browser initiates a server request. Dynamic pages are then created based on several conditions. | It happens when the browser possesses all the codes and the page in later changed according to the user's input. |
| Execution | The scripting process for the server side is done on remote computer and hence the response is comparatively slower than the client side one. | The scripting process of client server is executed on a local computer and thus the response is comparatively quicker when compared to server side scripting. |
| Connection to the database | It helps in connecting to the databases that are already present in the web server. | It does not connect to the databases that are on the web server. |
| Suitability | It is excellent for any area that requires loading of dynamic data. | It is excellent for any case which requires user interaction. |
| Languages | Languages used in server scripting are Ruby on Rails, PHP, ColdFusion, Python, ASP, Perl etc | Languages commonly used for client-side scripting are Javascript, HTML, CSS, VB script etc. |
| Access To Files | It has access to all the files present in the web server. | It has no access to all the files present in the web server. |
| Security | It is more secure than client side scripting as the server side scripts are usually hidden from the client end. | It is less secure because the scripts are usually not hidden from the client end. |